

COMP 8005

Assignment 1

Testing

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Test	Expected	Actual	Screenshot
No arguments	fail	fail	Test 1
-h	pass	pass	Test 2
-x	fail	fail	Test 3
Missing argument	fail	fail	Test 4
Negative port	fail	fail	Test 5
Too many arguments	fail	fail	Test 6
Shadow file doesn't exist	fail	fail	Test 7
User doesn't exist	fail	fail	Test 8
MD5 password	pass	pass	Test 9
bcrypt password	pass	pass	Test 10
SHA-256 password	pass	pass	Test 11
SHA-512 password	pass	pass	Test 12
yescrypt password	pass	pass	Test 13
Worker Invalid IP Address	fail	fail	Test 14

Tests

Test 1

```

PS C:\Users\winst\OneDrive\Desktop\COMP8005 A1\COMP8005-Password-Cracking> python3 ./controller.py
usage: controller.py [-h] -f FILE -u USER -p PORT
controller.py: error: the following arguments are required: -f/--file, -u/--user, -p/--port
PS C:\Users\winst\OneDrive\Desktop\COMP8005 A1\COMP8005-Password-Cracking>

```

Test 2

```
PS C:\Users\winst\OneDrive\Desktop\COMP8005 A1\COMP8005-Password-Cracking> python3 ./controller.py -h
usage: controller.py [-h] -f FILE -u USER -p PORT

Password Cracker Controller Script

options:
  -h, --help            show this help message and exit
  -f, --file FILE       Path to the shadow file
  -u, --user USER       Username of the password to crack
  -p, --port PORT       Port number to host on
PS C:\Users\winst\OneDrive\Desktop\COMP8005 A1\COMP8005-Password-Cracking>
```

Test 3

```
PS C:\Users\winst\OneDrive\Desktop\COMP8005 A1\COMP8005-Password-Cracking> python3 ./controller.py -x
usage: controller.py [-h] -f FILE -u USER -p PORT
controller.py: error: the following arguments are required: -f/--file, -u/--user, -p/--port
PS C:\Users\winst\OneDrive\Desktop\COMP8005 A1\COMP8005-Password-Cracking>
```

Test 4

```
PS C:\Users\winst\OneDrive\Desktop\COMP8005 A1\COMP8005-Password-Cracking> python3 ./controller.py -u user -p 8080
usage: controller.py [-h] -f FILE -u USER -p PORT
controller.py: error: the following arguments are required: -f/--file
PS C:\Users\winst\OneDrive\Desktop\COMP8005 A1\COMP8005-Password-Cracking>
```

Test 5

```
PS C:\Users\winst\OneDrive\Desktop\COMP8005 A1\COMP8005-Password-Cracking> python3 ./controller.py -f /etc/shadow -u user -p -8080
usage: controller.py [-h] -f FILE -u USER -p PORT
controller.py: error: argument -p/--port: -8080 is not a positive integer
PS C:\Users\winst\OneDrive\Desktop\COMP8005 A1\COMP8005-Password-Cracking>
```

Test 6

```
PS C:\Users\winst\OneDrive\Desktop\COMP8005 A1\COMP8005-Password-Cracking> python3 ./controller.py -f /etc/shadow -u user -p 8080 1
usage: controller.py [-h] -f FILE -u USER -p PORT
controller.py: error: unrecognized arguments: 1
PS C:\Users\winst\OneDrive\Desktop\COMP8005 A1\COMP8005-Password-Cracking>
```

Test 7

```
PS C:\Users\winst\OneDrive\Desktop\COMP8005 A1\COMP8005-Password-Cracking> python3 ./controller.py -f /fake_file -u user -p 8080
Error: Failed to open shadow file.
PS C:\Users\winst\OneDrive\Desktop\COMP8005 A1\COMP8005-Password-Cracking>
```

Test 8

```
PS C:\Users\winst\OneDrive\Desktop\COMP8005 A1\COMP8005-Password-Cracking> python3 ./controller.py -f /etc/shadow -u fake_user -p 8080
Error: Username not found in shadow file.
PS C:\Users\winst\OneDrive\Desktop\COMP8005 A1\COMP8005-Password-Cracking> |
```

Test 9

```
PS C:\Users\winst\OneDrive\Desktop\COMP8005 A1\COMP8005-Password-Cracking> python3 ./controller.py -f /etc/shadow -u abc -p 8080
Server running on dual-stack mode (IPv4 and IPv6).
Server is listening on port 8080 for incoming connections...
Connection received: ('::ffff:192.168.1.90', 39548, 0, 0)
Sent data to ('::ffff:192.168.1.90', 39548, 0, 0)
Received data from client: {'password': 'abc', 'crack_time': datetime.timedelta(microseconds=571732), 'dispatch_latency': datetime.timedelta(microseconds=98725)}
=====
Hash Algorithm: MD5
Password Found: abc
Controller Parsing Time: 0.007093 seconds
Dispatch Latency: 0.194731 seconds
Cracking Time: 0.571732 seconds
Return Latency: 0.205381 seconds
Total end-to-end Runtime: 3.561421 seconds
=====
```

Test 10

```
PS C:\Users\winst\OneDrive\Desktop\COMP8005 A1\COMP8005-Password-Cracking> python3 ./controller.py -f /etc/shadow -u abc -p 8080
Server running on dual-stack mode (IPv4 and IPv6).
Server is listening on port 8080 for incoming connections...
Connection received: ('::ffff:192.168.1.90', 56744, 0, 0)
Sent data to ('::ffff:192.168.1.90', 56744, 0, 0)
Received data from client: {'password': 'abc', 'crack_time': datetime.timedelta(seconds=9, microseconds=178933), 'dispatch_latency': datetime.timedelta(seconds=54, microseconds=539788)}
=====
Hash Algorithm: bcrypt
Password Found: abc
Controller Parsing Time: 0.007546 seconds
Dispatch Latency: 0.196824 seconds
Cracking Time: 9.178933 seconds
Return Latency: 0.203619 seconds
Total end-to-end Runtime: 12.165712 seconds
=====
```

Test 11

```
PS C:\Users\winst\OneDrive\Desktop\COMP8005 A1\COMP8005-Password-Cracking> python3 ./controller.py -f /etc/shadow -u abc -p 8080
Server running on dual-stack mode (IPv4 and IPv6).
Server is listening on port 8080 for incoming connections...
Connection received: ('::ffff:192.168.1.90', 37866, 0, 0)
Sent data to ('::ffff:192.168.1.90', 37866, 0, 0)
Received data from client: {'password': 'abc', 'crack_time': datetime.timedelta(seconds=7, microseconds=45302), 'dispatch_latency': datetime.timedelta(seconds=57, microseconds=668755)}
=====
Hash Algorithm: SHA-256
Password Found: abc
Controller Parsing Time: 0.00858 seconds
Dispatch Latency: 0.196985 seconds
Cracking Time: 7.045302 seconds
Return Latency: 0.204128 seconds
Total end-to-end Runtime: 8.56452 seconds
=====
```

Test 12

```
PS C:\Users\winst\OneDrive\Desktop\COMP8005 A1\COMP8005-Password-Cracking> python3 ./controller.py -f /etc/shadow -u abc -p 8080
Server running on dual-stack mode (IPv4 and IPv6).
Server is listening on port 8080 for incoming connections...
Connection received: ('::ffff:192.168.1.90', 43456, 0, 0)
Sent data to ('::ffff:192.168.1.90', 43456, 0, 0)
Received data from client: {'password': 'abc', 'crack_time': datetime.timedelta(seconds=9, microseconds=264756), 'dispatch_latency': 58, 4, 821880)}
=====
Hash Algorithm: SHA-512
Password Found: abc
Controller Parsing Time: 0.009052 seconds
Dispatch Latency: 0.196383 seconds
Cracking Time: 9.264756 seconds
Return Latency: 0.203022 seconds
Total end-to-end Runtime: 11.079179 seconds
=====
```

Test 13

```
PS C:\Users\winst\OneDrive\Desktop\COMP8005 A1\COMP8005-Password-Cracking> python3 ./controller.py -f /etc/shadow -u abc -p 8080
Server running on dual-stack mode (IPv4 and IPv6).
Server is listening on port 8080 for incoming connections...
Connection received: ('::ffff:192.168.1.90', 36394, 0, 0)
Sent data to ('::ffff:192.168.1.90', 36394, 0, 0)
Received data from client: {'password': 'abc', 'crack_time': datetime.timedelta(seconds=73, microseconds=82301), 'dispatch_latency': 59, 44, 753934)}
=====
Hash Algorithm: yescrypt
Password Found: abc
Controller Parsing Time: 0.007991 seconds
Dispatch Latency: 0.196443 seconds
Cracking Time: 73.082301 seconds
Return Latency: 0.204211 seconds
Total end-to-end Runtime: 74.996658 seconds
=====
```

Test 14

```
PS C:\Users\winst\OneDrive\Desktop\COMP8005 A1\COMP8005-Password-Cracking> python3 ./worker.py -c bad_ip -p 8080
Error: Invalid IP Address found.
PS C:\Users\winst\OneDrive\Desktop\COMP8005 A1\COMP8005-Password-Cracking>
```